

Web Development Practice Projects

Applied Problem Solving with HTML, CSS, and JavaScript

Prepared by: Yazdan Maqsood

1. Smart Theme Switcher

Build a webpage with a readable article and a "Toggle Theme" button. Write a JavaScript **function** that checks the current theme using an **if/else condition**. If the theme is light, switch the CSS classes to a dark mode layout (dark background, light text), and vice versa. Update the button text to reflect the current mode.

HTML

CSS

JS: Conditions, Functions

2. Dynamic Image Gallery

Create an empty CSS Grid container in your HTML. In your JavaScript file, create an **array** of 5 different image URLs. Write a **loop** (e.g., `for` or `forEach`) that iterates through the array, creates an `<img>` element for each URL, and appends it to the grid container.

HTML

CSS

JS: Loops, Arrays

3. Secure Registration Form

Design a signup form requesting an email and password. Create a validation **function** triggered on form submission. Use **string functions** (like `includes()`) to ensure the email contains an "@" symbol. Use **conditions** to check if the password length is at least 8 characters. Display styled error or success messages below the form.

HTML

CSS

JS: Conditions, String Functions

4. Word Counter and Highlighter

Build a text area where a user can type a paragraph, and an input field for a "Search Word". Write a script that uses **string functions** (like `split()`) to calculate and display the total number of words typed. When a word is entered in the search field, use `replaceAll()` to highlight occurrences of that word in a preview pane using a styled `<span>`.

HTML

CSS

JS: String Functions, Functions

5. Student Grades Dashboard

Create an **array of objects**, where each object represents a student (e.g., `{ name: 'Ali', score: 85 }`). Use the **array function** `filter()` to create a new array of students who scored above 50. Then, use `map()` to generate HTML list items for the passing students and display them on the page.

HTML

CSS

JS: Array Functions (map/filter)

6. Palindrome Checker

Create a simple UI with a text input and a "Check" button. Write a **function** that takes the input string, converts it to lowercase, and removes spaces. Use **string and array functions** (`split().reverse().join()`) or a **loop** to determine if the word is a palindrome (reads the same forwards and backwards). Output the result visually.

HTML

CSS

JS: Loops, String/Array Functions

7. Interactive Task Manager

Design a To-Do list where users can add tasks. Store tasks in a JavaScript **array**. Write a **function** to render the list to the DOM. Add a "Complete" button next to each task that uses an **array function** like `splice()` or `filter()` to remove the specific task from the array, then re-renders the updated list.

HTML

CSS

JS: Array Functions, Functions

8. Math Operator UI

Build a mini-calculator layout with two input fields and four buttons (+, -, *, /). Write a master **function** that accepts the two numbers and an operator string. Inside the function, use a switch statement or multiple **if/else conditions** to perform the correct calculation. Display the result prominently on the screen.

HTML

CSS

JS: Conditions, Functions

9. Shopping Cart Total Calculator

Set up an **array** containing price values of items in a shopping cart (e.g., [12.99, 5.50, 42.00]). Write a script utilizing the **array function** `reduce()` to calculate the total sum of all items. Apply a 10% discount if the total exceeds \$50 using **conditions**. Update a beautifully styled receipt container with the final price.

HTML

CSS

JS: Array Functions (reduce), Conditions

10. Quiz Application logic

Create an **array of objects** for a mini-quiz, containing properties for `question`, `correctAnswer`, and `userAnswer`. Write a **loop** that iterates over the questions and compares the user's answer to the correct one using **conditions**. Keep a running tally of the score, and use **string formatting** (Template Literals) to print a final results summary to the DOM.

HTML

CSS

JS: Loops, Conditions, Arrays, Strings